Result 1 (shifted perturbation value). [Topology foundations, constant real translation]
Let P be interlevel problem data. Let t be a real number. Let x be a point of P.space. Then

$$(P.map - t)\,(x) = P.\mathrm{map}(x) - t.$$

Result 2 (shifted perturbation bound). [Bendich, Edelsbrunner, Morozov, and Patel, extreme perturbations]
Let P be interlevel problem data. Let t and r be real numbers. Assume $-r \leq t$ (`lower`). Assume $t \leq r$ (`upper`). Then

$$(P.map - t) \text{ is a perturbation of } P \text{ bounded by } r.$$

Adapted from the cited source. Scope: uniform bound.

Result 3 (shifted perturbation pair). [Bendich, Edelsbrunner, Morozov, and Patel, shifted interlevel pairs]
Let P be interlevel problem data. Let t be a real number. Then

$$A(P, (P.map - t)) = T_t(P).$$

Adapted from the cited source.

Result 4 (shifted perturbations have certificates). [Bendich, Edelsbrunner, Morozov, and Patel, shifted perturbation family]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let t and r be real numbers. Assume P.interval.lower $\leq$ P.interval.upper (`order`). Assume $r \geq 0$ (`radius`). Assume $-r \leq t$ (`lower`). Assume $t \leq r$ (`upper`). Assume $X_r(P)$ is a nested pair (`nesting`). Then

$$(\text{there exists } C : \mathrm{CertifiedPerturbation},\ C.\mathrm{data.map} = (P.map - t))\,.$$

Adapted from the cited source. Scope: certificate completeness.

Result 5 (shifted image certificate independence). [Topology foundations, shifted image well-definedness]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let t and r be real numbers. Assume $X_r(P)$ is a nested pair (`nesting`). Let C and D be elements of CertifiedPerturbation. Assume C.data.map $= (P.map - t)$ (`first`). Assume D.data.map $= (P.map - t)$ (`second`). Then

$$I_t(P; H, r, C) = I_t(P; H, r, D).$$

Result 6 (well group lies in every shifted image). [Bendich, Edelsbrunner, Morozov, and Patel, well-group intersection definition]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let P be interlevel problem data. Let t and r be real numbers. Assume $X_r(P)$ is a nested pair (`nesting`). Let C be a bounded perturbation with a certified chain square. Assume C.data.map $= (P.map - t)$ (`represents`). Let x be an element of $H(X_r(P))$. Assume x belongs to $U_r(P; H)$ (`member`). Then

$$x \text{ belongs to } I_t(P; H, r, C).$$

Adapted from the cited source. Scope: one inclusion only.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.